Total No. of Questions : 8]

SEAT No. :

PA-2673

[Total No. of Pages : 2

[5927]-458
B.E (Mechanical/Automobile)
(Honors/Minors)
MODELLING AND SIMULATION OF EHV
(2019 Pattern) (Semester - VII) (402034MJ)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Figures to the right side indicate full marks.***
3) ***Draw the neat sketch wherever necessary.***

Q1) a) Explain Battery Management System along with its main features? **[4]**

b) Explain The Electric Machine Control System (EMCS) and The Stability Control System (SCS) with its important features? **[8]**

c) What is Battery/Cell Control System in EV? Explain with its functions and working principal? **[8]**

OR

Q2) a) Explain Torque and speed coupling in electric vehicle? **[4]**

b) Explain Electronic Control Unit of electric vehicle with its types and working principal? **[8]**

c) Explain the importance of Sensor Management and Integration for electric vehicle with functional and application point of view? **[8]**

Q3) a) Explain, "Placement of Motors" in electric vehicle system? **[8]**

b) Explain, "Battery and Motion Transmission Systems" in electric vehicle system? **[8]**

OR

Q4) a) List and explain with suitable sketches, the configurations of possible drivetrain systems in electric vehicle? **[8]**

b) Explain Propulsion and Power distribution system in electric vehicle with its main components? **[8]**

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Q5) a) What are Requirements of body structural system for a road vehicle? **[8]**

b) Explain causes for safety-related failure for lithium ion battery? **[8]**

OR

Q6) a) What do you mean by dynamics of motor vehicle? List out various force and performance parameters consider in it? **[8]**

b) Explain Chassis frame layout with suitable sketch? Also list out various types Loads on the Chassis frame? **[8]**

Q7) a) Explain Vehicle Structure design against Noise and Vibration exposure with suitable sketches? **[8]**

b) Explain the phases involved in Crashworthiness Design along with its important features? **[10]**

OR

Q8) a) Explain Noise Factors & Failure Modes in electric vehicle? **[10]**

b) What are Human Characteristics and Capabilities in ergonomics design of electric vehicle? **[8]**

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